

Q1 2021 (01 Sep Shift 2)

If the sum of the coefficients in the expansion of $(x + y)^n$ is 4096, then the greatest coefficient in the expansion is .

Q2 2021 (31 Aug Shift 2)

If the coefficient of $a^7 b^8$ in the expansion of $(a + 2b + 4ab)^{10}$ is $K \cdot 2^{16}$, then K is equal to .

Q3 2021 (31 Aug Shift 1)

If $\left(\frac{3^6}{4^4}\right)k$ is the term, independent of x, in the binomial expansion of $\left(\frac{x}{4} - \frac{12}{x^2}\right)^{12}$, then k is equal to .

Q4 2021 (27 Aug Shift 1)

$\sum_{k=0}^{20} ({}^{20}C_k)^2$ is equal to :

- (1) ${}^{40}C_{21}$
- (2) ${}^{40}C_{19}$
- (3) ${}^{40}C_{20}$
- (4) ${}^{41}C_{20}$

Q5 2021 (26 Aug Shift 2)

Let $\binom{n}{k}$ denotes nC_k and $\left[\begin{matrix} n \\ k \end{matrix} \right] = \begin{cases} \binom{n}{k}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$

If $A_k = \sum_{i=0}^9 \binom{9}{i} \left[\begin{matrix} 12 \\ 12 - k + i \end{matrix} \right] + \sum_{i=0}^8 \binom{8}{i} \left[\begin{matrix} 13 \\ 13 - k + i \end{matrix} \right]$

and $A_4 - A_3 = 190p$, then p is equal to :

Q6 2021 (26 Aug Shift 1)

If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1 + x)^{20}$, then the value of $\sum_{r=0}^{20} r^{220} C_r$ is equal to :

- (1) 420×2^{19}

(4) 420×2^{18}

Q6 (4)

Q1 (924)

$$(x + y)^n \Rightarrow 2^n = 4096$$

$$\Rightarrow 2^n = 2^{12}$$

$$n = 12$$

$$2^{10} = 1024 \times 2$$

$$2^{11} = 2048$$

$$2^{12} = 4096$$

$${}^{12}C_6 = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= 11 \times 3 \times 4 \times 7$$

$$= 924$$

Q2 (315)

$$\frac{10!}{\alpha! \beta! \gamma!} a^\alpha (2b)^\beta \cdot (4ab)^\gamma$$

$$\frac{10!}{\alpha! \beta! \gamma!} a^{\alpha+\gamma} \cdot b^{\beta+\gamma} \cdot 2^\beta \cdot 4^\gamma$$

$$\alpha + \beta + \gamma = 10 \dots (1)$$

$$\alpha + \gamma = 7 \dots (2)$$

$$\beta + \gamma = 8 \dots (3)$$

$$(2) + (3) - (1) \Rightarrow \gamma = 5$$

$$\alpha = 2$$

$$\beta = 3$$

$$\text{so coefficients} = \frac{10!}{2!3!5!} 2^3 \cdot 2^{10}$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{2 \times 3 \times 2 \times 5!} \times 2^{13}$$

$$= 315 \times 2^{16} \Rightarrow k = 315.$$

Q3 (55)

$$\left(\frac{x}{4} - \frac{12}{x^2}\right)^{12}$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{x}{4}\right)^{12-r} \left(\frac{12}{x^2}\right)^r$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{1}{4}\right)^{12-r} (12)^r \cdot (x)^{12-3r}$$

Term independent of $x \Rightarrow 12 - 3r = 0 \Rightarrow r = 4$

$$T_5 = (-1)^4 \cdot {}^{12}C_4 \left(\frac{1}{4}\right)^8 (12)^4 = \frac{3^6}{4^4} \cdot k$$

$$\Rightarrow k = 55$$

Q4 (3)

$\sum_{k=0}^{20} {}^{20}C_k \cdot {}^{20}C_{20-k}$ sum of suffix is const. so summation will be ${}^{40}C_{20}$

Q5 (49)

$$A_k = \sum_{i=0}^9 {}^9C_i {}^{12}C_{k-i} + \sum_{i=0}^8 {}^8C_i {}^{13}C_{k-i}$$

$$A_k = {}^{21}C_k + {}^{21}C_k = 2 \cdot {}^{21}C_k$$

$$A_4 - A_3 = 2 ({}^{21}C_4 - {}^{21}C_3) = 2(5985 - 1330)$$

$$190p = 2(5985 - 1330) \Rightarrow p = 49$$

Q6 (4)

$$\sum_{r=0}^{20} r^2 \cdot {}^{20}C_r$$

$$\sum (4(r-1) + r) \cdot {}^{20}C_r$$

$$\sum r(r-1) \cdot \frac{20 \times 19}{r(r-1)} \cdot {}^{18}C_{r-2} + r \cdot \frac{20}{r} \cdot \sum {}^{19}C_{r-1}$$

$$\Rightarrow 20 \times 19 \cdot 2^{18} + 20 \cdot 2^{19}$$

$$\Rightarrow 420 \times 2^{18}$$